

# 2016 APMCM Problem B

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## The Influence of Chemical Element on Properties of Deformed Steel Bar

Hot-rolled ribbed bar is commonly known as deformed steel bar, it's mainly used for skeleton of reinforced concrete component and it requires certain mechanical strength, bending and deformation properties, fabrication weldability in use. The chemical composition in steel is the basic element that influences the final structure property of hot rolled steel. Most deformed steel bar adopts microalloying method, that is to add expensive microelement (such as Mn alloy material, V alloy material, etc.) into steel, adjust the composition proportion and to improve structure property. A good composition design can guarantee the properties and control the production cost effectively at the same time. The element Cr in steel can significantly increase the strength, hardness and wear resistance. When the company uses mine rich in Cr, the Cr content in liquid iron will significantly increase. Therefore there is a question: when Cr content increases, can we reduce alloy material amount in order to control the cost?

There is no clear dependency between the properties of deformed steel bar like tensile strength, yield strength and percentage elongation after fracture, elements like C, Mn, S, P, Si, Cr, Mo, Cu, Ni, Al, V and other influence factors, so please apply modern mathematical and statistical methods in accordance with data in attachment 1 to model the influence of deformed steel bar properties on chemical element and optimize the composition. Specific requirements are as follows:

- (1) Analyze the main factors, secondary factors that influence properties of deformed steel bar like yield strength, tensile strength and percentage elongation after fracture, and analyze the correlation between these factors.
- (2) Model the influence rule between deformed steel bar properties and chemical elements like C, Mn, Cr, V, Ni.
- (3) Research the reduction of alloy material like Mn and V in allowance range of deformed steel bar properties by Cr content increase, present the content modification scope of Mn, V, etc, and design composition optimization scheme.

Attachment 1: the production data in the past two years of a certain enterprise. Among them: the specification is classified according to diameter of deformed steel bar, there are only two kinds of products in the table; each element content is shown in percentage; unit of yield strength, tensile strength is MPa, percentage elongation after fracture is shown in percentage.